

## Initial Project Planning Template

|               |                                                             |
|---------------|-------------------------------------------------------------|
| Date          | 4 July 2026                                                 |
| Team ID       | XXXXXX                                                      |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 5 Marks                                                     |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                                                           | Story Points | Priority | Team Members                     | Sprint Start Date | Sprint End Date (Planned) |
|----------|-------------------------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------|--------------|----------|----------------------------------|-------------------|---------------------------|
| Sprint-1 | Define Problem & Planning     | USN-1             | As a project team member, I want to define the business problem and project workflow so that the project objectives are clearly understood. | 2            | High     | Badikala Pranathi, Molaga Manasa | 25-06-2026        | 25-06-2026                |
| Sprint-1 | Data Collection               | USN-2             | As a data analyst, I want to collect and import the agricultural dataset so that it can be used for analysis and model training.            | 3            | High     | Molaga Manasa                    | 25-06-2026        | 26-06-2026                |
| Sprint-2 | Data Analysis                 | USN-3             | As a data analyst, I want to analyze the dataset using univariate, bivariate, and                                                           | 5            | High     | Kotturu Malik Basha              | 27-06-2026        | 28-06-2026                |

|          |                      |       |                                                                                                                                                 |   |        |                 |            |            |
|----------|----------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|-----------------|------------|------------|
|          |                      |       | multivariate analysis so that important patterns can be identified.                                                                             |   |        |                 |            |            |
| Sprint-2 | Data Preprocessing   | USN-4 | As a machine learning engineer, I want to clean and preprocess the dataset so that the model receives high-quality data for training.           | 5 | Medium | Madapogu Eshwar | 28-06-2026 | 30-06-2026 |
| Sprint-3 | Model Building       | USN-5 | As a machine learning engineer, I want to train and evaluate multiple machine learning models so that the most accurate model is selected.      | 8 | High   | Madapogu Eshwar | 01-07-2026 | 02-07-2026 |
| Sprint-3 | Crop Prediction      | USN-6 | As a farmer, I want to receive the best crop recommendation based on soil and environmental parameters so that I can improve crop yield.        | 5 | High   | Victor Joshua   | 02-07-2026 | 03-07-2026 |
| Sprint-4 | Web Application      | USN-7 | As a farmer, I want to enter soil and climate details through a simple web interface so that I can easily obtain crop recommendations.          | 5 | High   | Victor Joshua   | 03-07-2026 | 04-07-2026 |
| Sprint-4 | Testing & Deployment | USN-8 | As a user, I want the application to provide accurate and reliable crop recommendations so that I can confidently use it for farming decisions. | 3 | High   | Victor Joshua   | 04-07-2026 | 05-07-2026 |